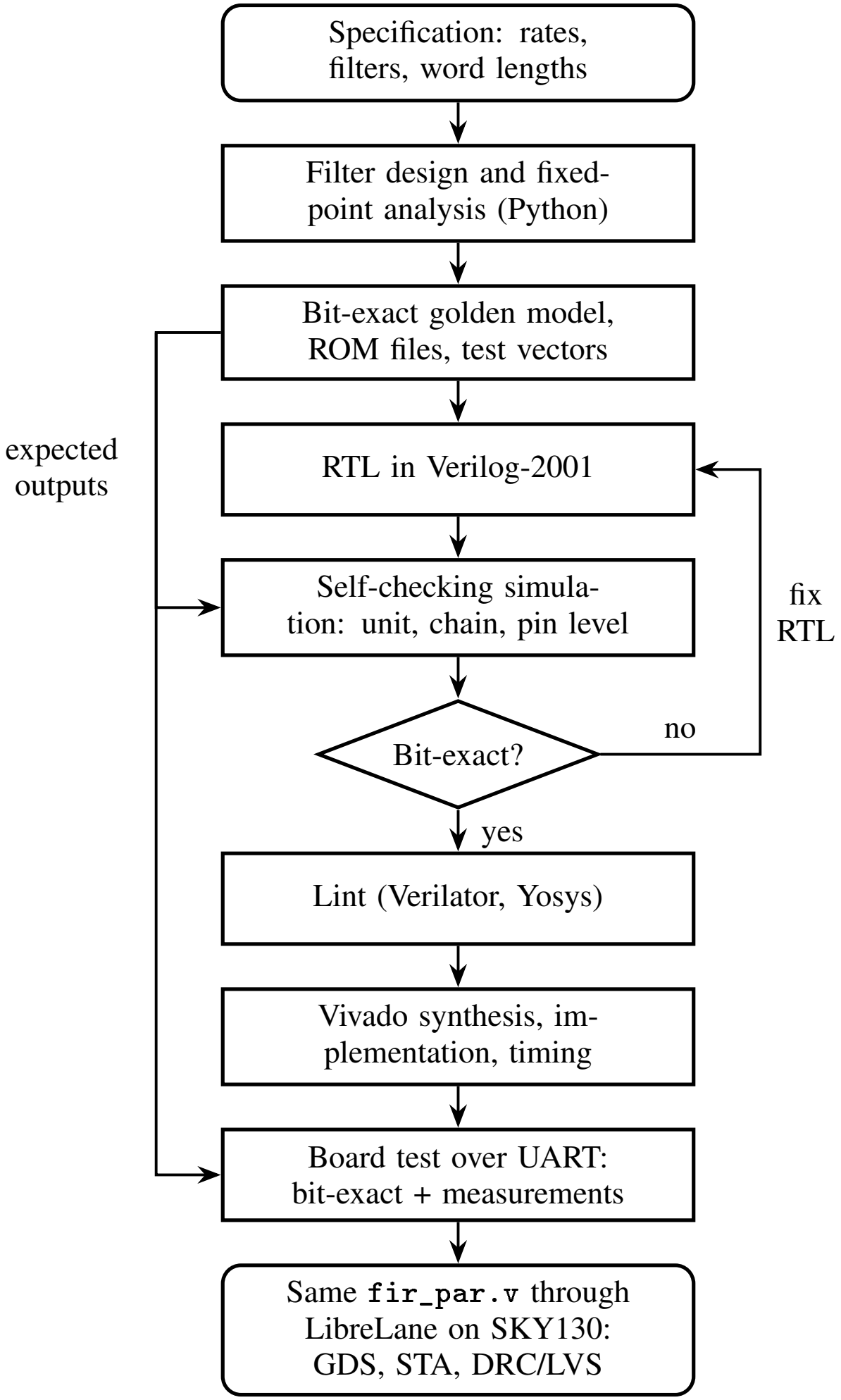
Specification: rates, filters, word lengths
Filter design and fixed-point analysis (Python)
Bit-exact golden model, ROM files, test vectors
expected outputs
RTL in Verilog-2001
Self-checking simulation: unit, chain, pin level
fix RTL
Bit-exact?
no
yes
Lint (Verilator, Yosys)
Vivado synthesis, implementation, timing
Board test over UART: bit-exact + measurements
Same fir_par.v through LibreLane on SKY130: GDS, STA, DRC/LVS